

Summary Tables

Systematic Mapping and Benchmarking of Sign Language Datasets

The following five tables present structured summaries of the 30 sign language datasets included in this systematic mapping study. Each table addresses one analytical dimension. Tables 1 to 3 are presented in landscape orientation to accommodate the number of columns. Tables 4 and 5 are presented in portrait orientation. Across all tables, "—" indicates that the information was not reported in the source publication. In Tables 2 and 3, "Yes" indicates the feature is present, "No" indicates it is absent, and "Partial" indicates that it is partially present or only implied by the source.

Contents

Table 1. Scale and scope of the 30 included datasets.

Table 2. Modality coverage across the 30 datasets.

Table 3. Annotation and benchmarking characteristics.

Table 4. Recording conditions.

Table 5. Signer demographics.

Table 1. Scale and scope of the 30 sign language datasets included in the systematic mapping study.

Dataset	Year	Type	#Sign Classes	#Video Samples	Total Hours	#Signers
How2Sign	2021	Continuous	16,000 words	2,653	82.08 h	11
MS-ASL	2018	Isolated	1000	25513	~24.0 h	222
WLASL (Word-Level American Sign Language)	2020	Isolated	2000	21083	~14.0 h	119
YouTube-ASL	2023	Continuous	60,000	11093	984.0 h	2519
OpenASL	2022	Continuous	33,549	98417	288.0 h	220
BBC-Oxford British Sign Language (BOBSL)	2021	Continuous	2281	1962	~1467.0 h	39
RWTH-PHOENIX-Weather 2014	2014	Continuous	1,558	645	10.73 h	9
SIGNUM Database	2010	Mixed	450	33210	55.3 h	25
AUTSL (Ankara University Turkish Sign Language Dataset)	2020	Isolated	226	38336	—	43
BosphorusSign22k	2020	Isolated	744	22542	~19.0 h	6
Slovo: Russian Sign Language Dataset	2023	Isolated	1000	20000	19.81 h	194
Corpus of Finnish Sign Language (Corpus FinSL)	2020	Continuous	107000	343	14.37 h	21
RoCoISLR (Romanian Isolated Sign Language Recognition)	2025	Isolated	5892	9141	—	25
BdSLW60 (Bangla Sign Language Word-Level 60)	2024	Isolated	60	9307	—	18
INCLUDE (Indian Lexicon Sign Language Dataset)	2020	Isolated	263	4287	~3.0 h	7
CE-CNSL (Complex Environment - Chinese Sign Language dataset)	2024	Continuous	3515	5988	10.52 h	12
CSL-Daily	2021	Continuous	2000	20654	23.27 h	10
SL-ReDu GSL corpus	2023	Mixed	369	30116	35.93 h	21
SMILE Swiss German Sign Language Dataset	2018	Isolated	100	9000	—	30
MM-WLAuslan (Multi-View Multi-Modal Word-Level Australian Sign Language Recognition Dataset)	2024	Isolated	3215	282900	—	73
LSA64: An Argentinian Sign Language Dataset	2023	Isolated	64	3200	—	10

Dataset	Year	Type	#Sign Classes	#Video Samples	Total Hours	#Signers
AzSLD (Azerbaijani Sign Language Dataset)	2025	Mixed	719	30312	65.2 h	43
King Saud University Saudi-SSL (KSU-SSL database)	2023	Mixed	293	145035	495.0 h	33
Alabib-65	2023	Isolated	65	6238	—	41
Mediapi-RGB	2024	Continuous	35599	1230	86.0 h	>10
SWL-LSE (SignaMed Word-Level Lengua de Signos Española)	2024	Isolated	300	8000	—	124
SignIT	2025	Isolated	94	644	3.33 h	45
CASL-W60	2025	Isolated	60	5889	—	19
KSL-Guide	2021	Mixed	2000	521,000	—	72
LIBRAS-UFOP	2021	Isolated	56	3040	—	5

***Note.** The #Sign Classes column reports the number of distinct sign glosses or word classes in each dataset. The #Video Samples column reports the total number of video clips or sequences. Total hours reflects the cumulative duration of video footage as reported by the original authors. #Signers reports the total number of individual signing participants. All values are extracted from the source publications; "—" indicates that the value was not explicitly reported.*

Table 2. Modality coverage across the 30 included sign language datasets.

Dataset	RGB Video	Depth Video	Skeleton / Pose	Multi-view	Audio
How2Sign	Yes	Yes	Yes	Yes	Yes
MS-ASL	Yes	No	Yes	No	No
WLASL (Word-Level American Sign Language)	Yes	No	Yes	No	No
YouTube-ASL	Yes	No	No	No	No
OpenASL	Yes	No	No	No	No
BBC-Oxford British Sign Language (BOBSL)	Yes	No	Yes	No	No
RWTH-PHOENIX-Weather 2014	Yes	No	No	No	Yes
SIGNUM Database	Yes	No	No	No	No
AUTSL (Ankara University Turkish Sign Language Dataset)	Yes	Yes	Yes	No	No
BosphorusSign22k	Yes	Yes	Yes	No	No
Slovo: Russian Sign Language Dataset	Yes	No	No	No	No
Corpus of Finnish Sign Language (Corpus FinSL)	Yes	No	No	No	No
RoCoISLR (Romanian Isolated Sign Language Recognition)	Yes	No	No	No	No
BdSLW60 (Bangla Sign Language Word-Level 60)	Yes	No	Yes	Partial	No
INCLUDE (Indian Lexicon Sign Language Dataset)	Yes	No	Yes	No	No
CE-CNSL (Complex Environment - Chinese Sign Language dataset)	Yes	No	No	No	No
CSL-Daily	Yes	No	No	No	No
SL-ReDu GSL corpus	Yes	Yes	No	No	No
SMILE Swiss German Sign Language Dataset	Yes	Yes	Yes	Yes	No
MM-WLAuslan (Multi-View Multi-Modal Word-Level Australian Sign Language Recognition Dataset)	Yes	Yes	Yes	Yes	No
LSA64: An Argentinian Sign Language Dataset	Yes	No	Yes	No	No
AzSLD (Azerbaijani Sign Language Dataset)	Yes	No	No	Yes	No
King Saud University Saudi-SSL (KSU-SSL database)	Yes	No	Yes	No	No
Alabib-65	Yes	No	No	No	No

Dataset	RGB Video	Depth Video	Skeleton / Pose	Multi-view	Audio
Mediapi-RGB	Yes	No	No	No	Partial
SWL-LSE (SignaMed Word-Level Lengua de Signos Española)	Partial	No	Yes	No	No
SignIT	Yes	No	Yes	No	No
CASL-W60	Yes	No	Yes	No	No
KSL-Guide	Yes	No	Yes	Yes	No
LIBRAS-UFOP	Yes	Yes	Yes	No	No

***Note.** "Yes" indicates that the modality is included as part of the released dataset. "No" indicates that the modality is explicitly absent or that the authors explicitly avoided it. "Partial" indicates that the modality is present for a subset of the data or is derived rather than directly captured. Skeleton / Pose data refers to 2D or 3D body, hand, and face keypoints extracted using tools such as OpenPose, MediaPipe, or AlphaPose.*

Table 3. Annotation and benchmarking characteristics of the 30 included sign language datasets.

Dataset	Gloss Labels	Sentence Alignment	Non-Manual Annotation	Signer-Indep. Split	Baseline Results
How2Sign	Yes	Yes	Yes	Yes	Yes
MS-ASL	Yes	No	No	Yes	Yes
WLASL (Word-Level American Sign Language)	Yes	No	No	Yes	Yes
YouTube-ASL	No	Yes	No	No	Yes
OpenASL	No	Yes	No	No	Yes
BBC-Oxford British Sign Language (BOBSL)	Yes	Yes	Partial	Yes	Yes
RWTH-PHOENIX-Weather 2014	Yes	Yes	No	Yes	Yes
SIGNUM Database	Yes	Yes	No	Yes	Yes
AUTSL (Ankara University Turkish Sign Language Dataset)	Yes	No	No	Yes	Yes
BosphorusSign22k	Yes	No	Partial	Yes	Yes
Slovo: Russian Sign Language Dataset	Yes	Yes	No	No	Yes
Corpus of Finnish Sign Language (Corpus FinSL)	Yes	Yes	Partial	No	No
RoCoISLR (Romanian Isolated Sign Language Recognition)	Yes	No	No	No	Yes
BdSLW60 (Bangla Sign Language Word-Level 60)	Yes	No	Partial	Yes	Yes
INCLUDE (Indian Lexicon Sign Language Dataset)	Yes	No	No	No	Yes
CE-CNSL (Complex Environment - Chinese Sign Language dataset)	Yes	Yes	Partial	No	Yes
CSL-Daily	Yes	Yes	No	No	Yes
SL-ReDu GSL corpus	Yes	Yes	Yes	—	No
SMILE Swiss German Sign Language Dataset	Yes	No	No	Yes	No

Dataset	Gloss Labels	Sentence Alignment	Non-Manual Annotation	Signer-Indep. Split	Baseline Results
MM-WLAuslan (Multi-View Multi-Modal Word-Level Australian Sign Language Recognition Dataset)	Yes	No	No	Yes	Yes
LSA64: An Argentinian Sign Language Dataset	Yes	No	No	Yes	Yes
AzSLD (Azerbaijani Sign Language Dataset)	Yes	Yes	No	No	Yes
King Saud University Saudi-SSL (KSU-SSL database)	Yes	No	No	No	Yes
Alabib-65	Yes	No	No	Yes	Yes
Mediapi-RGB	No	Yes	No	No	Yes
SWL-LSE (SignaMed Word-Level Lengua de Signos Española)	Yes	Yes	No	Yes	Yes
SignIT	Yes	No	Partial	—	Yes
CASL-W60	Yes	No	No	Yes	Yes
KSL-Guide	Yes	Yes	No	No	Yes
LIBRAS-UFOP	Yes	No	Yes	Yes	Yes

Note. Gloss Labels indicates whether the dataset provides word-level sign annotations aligned with video segments. Sentence Alignment indicates whether signed utterances are aligned with spoken or written language translations. Non-Manual Annotation indicates whether facial expressions, mouthing, or other non-manual features are explicitly annotated. Signer-Indep. Split indicates whether the evaluation protocol excludes training-set signers from the test set. Baseline Results indicates whether the source publication reports model performance results on the dataset.

Table 4. Recording conditions of the 30 included sign language datasets.

Dataset	Environment	Background	Lighting	Camera Type	Resolution	Frame Rate (fps)
How2Sign	Studio	Simple	—	HD + RGB-D	1280 x 720	30
MS-ASL	Natural	Complex	Varied	Unconstrained	Standardized	Varied
WLASL (Word-Level American Sign Language)	Online	Simple	Varied	Monocular	Varied	Varied
YouTube-ASL	Online	Real-world	Varied	Varied	480 x 360	Varied
OpenASL	Online	Real-world	Varied	Varied	224 x 224	25
BBC-Oxford British Sign Language (BOBSL)	Broadcast	Simple	Controlled	Broadcast	444 x 444	25
RWTH-PHOENIX-Weat her 2014	Studio	Controlled	Controlled	—	210 x 260	25
SIGNUM Database	Lab	Simple	Controlled	Machine vision	776 x 578	30
AUTSL (Ankara University Turkish Sign Language Dataset)	Mixed	Complex	Varied	Kinect v2	512 x 512	—
BosphorusSign22k	Studio	Simple	—	Kinect v2	RGB: 1920 x 1080 Depth: 512 x 424	30
Slovo: Russian Sign Language Dataset	Natural	Varied	Varied	Laptop/Smart phone	1280x720	30
Corpus of Finnish Sign Language (Corpus FinSL)	Studio	—	—	—	—	—
RoCoSLR (Romanian Isolated Sign Language Recognition)	Mixed	Simple	—	—	224 x 224	25
BdSLW60 (Bangla Sign Language Word-Level 60)	Natural	Complex	—	Webcam	—	Varied
INCLUDE (Indian Lexicon Sign Language Dataset)	Natural	Complex	Natural	—	1920 x 1080	25
CE-CNSL (Complex Environment - Chinese Sign Language dataset)	Natural	Complex	Natural	Smartphone	Varied	Varied
CSL-Daily	Lab	—	—	—	1920 x 1080	30
SL-ReDu GSL corpus	Studio	Simple	Controlled	Sony + RealSense	1920 x 1080	RGB: 60Hz Depth: 30Hz
SMILE Swiss German Sign Language Dataset	Studio	Simple	Controlled	Kinect + GoPro + Webcams	Varied	Varied
MM-WLAuslan (Multi-View Multi-Modal Word-Level Australian Sign Language Recognition Dataset)	Studio	Simple	Controlled	Kinect + RealSense	—	—

Dataset	Environment	Background	Lighting	Camera Type	Resolution	Frame Rate (fps)
LSA64: An Argentinian Sign Language Dataset	Mixed	Simple	Mixed	Sony	—	—
AzSLD (Azerbaijani Sign Language Dataset)	Mixed	Simple	Controlled	HD RGB cameras	224 x 224	30
King Saud University Saudi-SSL (KSU-SSL database)	Studio	Simple	Controlled	Basler + IR + Mobile	1920 x 1200	60
Alabib-65	Mixed	Complex	Varied	iPad	Varied	30
Mediapi-RGB	Natural	Complex	Professional	Professional	Varied	Varied
SWL-LSE (SignaMed Word-Level Lengua de Signos Española)	Natural	Complex	Varied	Webcam + Mobile	Varied	Varied
SignIT	Natural	Complex	Varied	Web camera	Varied	Varied
CASL-W60	Mixed	Complex	Varied	Webcam	224 x 224	30
KSL-Guide	Mixed	Mixed	Varied	Multi-camera	1920 x 1080	30
LIBRAS-UFOP	Mixed	Mixed	Controlled	Kinect	—	30

Note. Environment classifies the recording setting as Studio (controlled laboratory or studio setup), Natural (real-world or in-the-wild recording), Broadcast (TV or online broadcast footage), Online (footage sourced from online platforms such as YouTube), Mixed (combination of conditions), or Lab (controlled laboratory environment). Background and Lighting classifications are derived from descriptions in the source publications.

Table 5. Signer demographics reported across the 30 included sign language datasets.

Dataset	#Signers	Deaf / Hearing / CODA	Gender Distribution	Age Range
How2Sign	11	4Deaf, 5Hearing, 2Hard-of-hearing	5F / 6M	—
MS-ASL	222	Students / Teachers	—	—
WLASL (Word-Level American Sign Language)	119	Native / Expert	—	—
YouTube-ASL	>2519	All skill levels	—	—
OpenASL	>220	Native	48.2% F / 42.7% M / 9.1% other	Child-older adult
BBC-Oxford British Sign Language (BOBSL)	39	Hearing, deaf interpreters	—	—
RWTH-PHOENIX-Weather 2014	9	Hearing interpreters	—	—
SIGNUM Database	25	23Deaf , 2Hearing-impaired	13F / 12M	21–50
AUTSL (Ankara University Turkish Sign Language Dataset)	43	1Deaf, 1CODA, 6TSL instructors, 3TSL translators, 25TSL course students, and 7trained signers	33F / 10M	—
BosphorusSign22k	6	Native(deaf)	—	—
Slovo: Russian Sign Language Dataset	194	Crowdworkers	—	—
Corpus of Finnish Sign Language (Corpus FinSL)	21	Native(deaf)	—	—
RoCoISLR (Romanian Isolated Sign Language Recognition)	25	Hearing / Mixed	—	—
BdSLW60 (Bangla Sign Language Word-Level 60)	18	Hearing volunteers	—	—
INCLUDE (Indian Lexicon Sign Language Dataset)	7	Deaf students	—	—
CE-CNSL (Complex Environment - Chinese Sign Language dataset)	12	2Hearing-impaired, 10Interpreters	8F / 4M	20–45
CSL-Daily	10	Native (Deaf community)	—	—
SL-ReDu GSL corpus	21	Deaf (L1), Bilingual	11F / 10M	19–52
SMILE Swiss German Sign Language Dataset	30	L1 signers + L2 learners	—	Adults
MM-WLAuslan (Multi-View Multi-Modal Word-Level Australian Sign Language Recognition Dataset)	73	Deaf, Experts, Volunteers	Volunteers: 26.0% F / 39.7% M Deaf individuals: 12.3% F / 8.2% M	—
LSA64: An Argentinian Sign Language Dataset	10	Non-expert subjects	—	—
AzSLD (Azerbaijani Sign Language Dataset)	43	Native, Interpreters, Users	Diverse	—

Dataset	#Signers	Deaf / Hearing / CODA	Gender Distribution	Age Range
King Saud University Saudi-SSL (KSU-SSL database)	33	Deaf, Hard-of-hearing, Hearing volunteers	—	—
Alabib-65	41	38Deaf, 3Hard-of-hearing	20F / 21M	6–18
Mediapi-RGB	>10	Deaf journalists	—	—
SWL-LSE (SignaMed Word-Level Lengua de Signos Española)	124	Deaf, Interpreters, LSE Learners	—	—
SignIT	45	—	—	—
CASL-W60	19	Trained volunteers	1F / 18M	22–35
KSL-Guide	72	Native KSL signers	35F / 17M	—
LIBRAS-UFOP	5	Trained volunteers	3F / 2M	—

***Note.** The #Signers column reports the total number of signing participants as stated by the authors. Deaf / Hearing / CODA reports the signer community composition where disclosed; CODA refers to Children of Deaf Adults. Gender Distribution is reported as F (female) and M (male) where provided. Age Range is extracted from the source text; "—" indicates that the information was not reported.*